# Learning to Listen:

## A Cost-Aware Ablation Study of a Perch-Distilled Sound Event Detection Network for Multi-Taxon Passive Acoustic Monitoring

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**Abstract**

We want the cheapest species classifier that still works well enough to run continuously on site for passive acoustic monitoring of the Brazilian Pantanal. We study a ResNet-18 sound event detection (SED) network that classifies 234 bird, amphibian, insect, mammal and reptile classes from 5 s audio windows, using the BirdCLEF 2026 data. The full model combines several components: 3-channel log-mel/delta/delta-delta input, a GeM-frequency + attention-over-time SED head, distillation of embeddings from Google's Perch v2 model with a stop-gradient on the classifier, Perch-localised pseudo-labels for focal recordings and unlabelled soundscapes, mixup, SpecAugment and rare-class upsampling. Each component was removed one at a time (11 training ablations). We also evaluated two baselines (a plain pooled ResNet-18 and a frozen Perch attention probe) and six post-processing variants that need no retraining. Every variant was scored on 13 metrics and benchmarked for inference cost.

On 3-fold, pooled out-of-fold evaluation over labelled soundscape windows (site S22 excluded), the full model reaches a macro ROC-AUC of 0.7142. Two simpler models score higher: the plain ResNet-18 baseline reaches 0.7571, and removing the pseudo-labelled soundscapes reaches 0.7742 while cutting expected calibration error from 0.184 to 0.003. Thirteen of 21 pre-registered hypotheses were supported. The ones that failed were about the expensive ideas: distillation, pseudo-labelling, GeM pooling, and the Perch teacher beating the student on rare species. Every CNN variant costs the same to run (11.7 M parameters, 5.9 GFLOPs, about 82 ms per window on one CPU thread, 61× real time), so component choices are free at inference and should be judged on quality alone. Three post-processing steps together (a fold-safe site/hour prior, temporal smoothing and taxonomy smoothing) raise macro ROC-AUC from 0.7142 to 0.8407 at essentially no inference cost. We report every number the study produced, give file-level bootstrap uncertainty, and discuss why the evaluation (16 soundscape files, 44 scorable classes, a 10-epoch budget) limits how far the rankings can be trusted.

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# 1 Introduction

Biodiversity monitoring increasingly relies on *passive acoustic monitoring* (PAM): autonomous recorders left in the field capture the calls of birds, frogs, insects and mammals around the clock. The bottleneck is analysis. A single recorder produces about 17,000 five-second windows per day, far more than any expert can label. The Pantanal of Brazil, the world's largest tropical wetland (over 150,000 $km^2$, with more than 650 bird species), is a prime example: it is vast, seasonally flooded, under threat from fire and land-use change, and largely unmonitored.

For a monitoring station the right question is not "what is the most accurate model?" but *"what is the cheapest model that is still good enough to run 24/7 on site?"* A model that needs a data-centre GPU cannot run on a solar-powered field box. This study is built around that question. Every variant is judged on **quality** (13 metrics covering ranking, detection, calibration, temporal stability and per-taxon performance) and on **cost** (parameters, GFLOPs, and CPU and GPU latency per prediction). A component earns its place only if its quality gain is worth its cost.

**Contributions.**

1. A cost-aware pipeline: a ResNet-18 student with an attention SED head, distilled offline from the Perch v2 bioacoustics model, with Perch also used to pseudo-label every training window (Section 3).
2. A controlled study of 20 variants: 2 baselines, the full model, 11 one-change training ablations and 6 post-hoc variants. Each tests a hypothesis written down before training (Section 3.6).
3. A complete report of every metric, per-fold selection statistic, learning curve, inference and training cost, and attention diagnostic produced by the study, plus file-level bootstrap confidence intervals that we added (Section 4).
4. An interpretation of what the evidence supports, what it does not, and which configuration we would deploy (Sections 5 and 7).

**Attribution.** The SED head, the Perch-distillation scheme with stop-gradient, the pseudo-labelling system and the site/hour priors are adapted from the "Model_1" and "Model_22" sections of the public Kaggle notebook *BirdCLEF 2026 EoS.9* by nina2025 [1]. That notebook credits tuckerarrants' distilled-SED and waveform-cache datasets and the BirdCLEF 2025 first-place SED design by nikitababich. Our changes are a ResNet-18 backbone in place of EfficientNet-B0 (with the global pool removed so the head sees the time axis), 3-channel stacked inputs, offline Perch pseudo-labels and embeddings, precomputed spectrograms, a constant learning rate, the ablation switches, the evaluation and cost harness, and this analysis.

# 2 Data

## 2.1 Sources

We use the BirdCLEF 2026 competition data [2]:

- **Focal recordings** (`train_audio`, `train.csv`): **35,549** clips from Xeno-canto and iNaturalist, each with a primary label and optional secondary labels. Secondary labels are present for **4,372** files. Clips are split into non-overlapping 5 s windows, giving **152,198 focal windows**.

- **Labelled soundscapes**: 66 one-minute field recordings, i.e. **792 windows** of 5 s (12 per file). **739** windows carry at least one label, with **3,122** positive (window, species) pairs covering **75 species**. File names encode the recording site and time, e.g. `BC2026_Train_0001_S08_20250606_030007.ogg`.
- **Unlabelled soundscapes**: **24,000** windows (2,000 one-minute files) used only through pseudo-labels.
- **Taxonomy**: **234 classes** in the competition's column order, each with a scientific name and taxonomic class (Aves, Amphibia, Insecta, Mammalia, Reptilia). Insect "sonotypes" (labels such as `47158son10`) are acoustic morphotypes without a species name.

## 2.2 Class imbalance

Figure 1 shows the number of focal clips per class for all 234 classes. **28 classes have no focal clips at all**: they can be learned only from soundscapes. **64 classes** (including those 28) have fewer than 20 clips and are called *rare*. 46 have 20–99, and **124** have at least 100 and are called *common*. The median is 104.5 clips per class and the maximum 499. The pile-up in the top bin suggests an upstream cap near 500 clips per class.

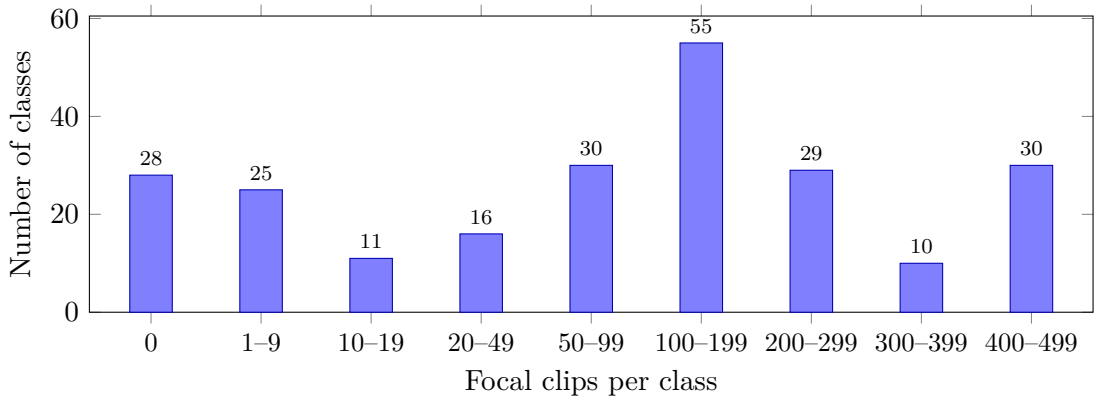



Figure 1: Focal clips per class for the 234 BirdCLEF 2026 classes (from `train_counts` in the out-of-fold files). The three left bars (64 classes) are the "rare" group; classes at or above 100 clips (124) are the "common" group.

Figure 2 is an earlier view of the same long tail, computed on the project's curated metadata table (169 labelled classes, counting primary and secondary labels).

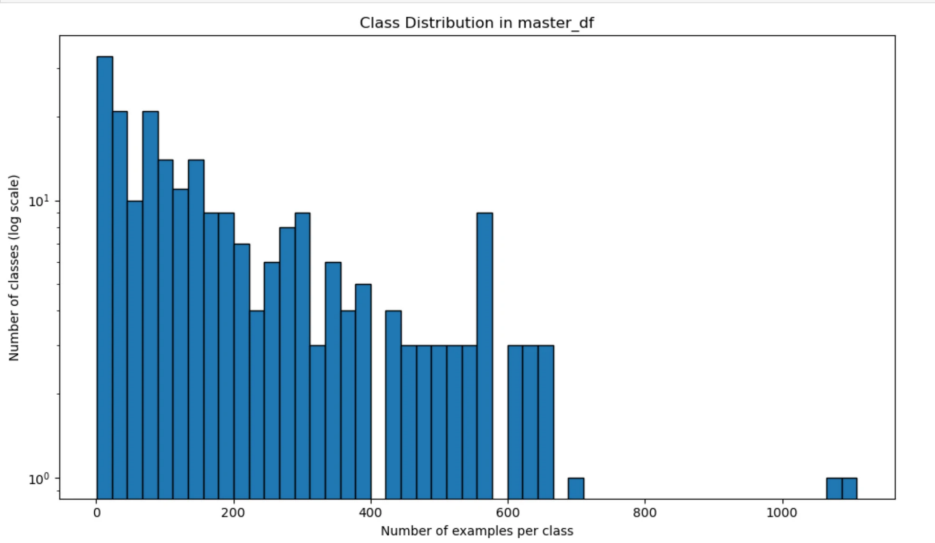



Figure 2: Log-scaled histogram of examples per class in the project's curated metadata table (169 classes). Most classes have only a few dozen examples and a handful have several hundred.

### 2.3 Cross-validation folds

Focal clips are split with `StratifiedKFold` (5 folds, stratified by primary label, seed 42), giving 7,110 clips in each of folds 0–3 and 7,109 in fold 4. Labelled soundscape files are split with `GroupKFold` (5 folds, grouped by file), giving 168 windows in fold 0 and 156 in each of folds 1–4. Because grouping is by file and not by site, the same site can appear in both training and validation folds. Pseudo-label priors for fold $k$ are fit without fold-$k$ files (Section 3.3). To fit the compute budget, **only folds 0, 1 and 2 were trained** for every variant.

### 2.4 Evaluation set

Following the competition notebook, site **S22** is excluded from scoring because of known label noise (480 of the 792 soundscape windows; 312 are non-S22). Predictions exist only for the validation windows of folds 0–2 (480 windows), so the **evaluation set is the 192 non-S22 windows of folds 0–2**:

- 16 soundscape files from 8 sites (S03, S08, S09, S13, S15, S18, S19, S23);
- 580 positive (window, class) pairs, i.e. 3.02 labels per window;
- **44 scorable classes** (at least one positive and one negative), with a median of 12 positive windows per class (minimum 1, maximum 60) and a mean prevalence of 0.069;
- 26 of the 44 scorable classes are rare ($<$20 focal clips) and only 8 are common ($\geq$100).

This set is small. Section 4.3 quantifies what that means for uncertainty.

# 3 Methods

## 3.1 Input features

Each 5 s window (160,000 samples at 32 kHz) is converted to a log-mel spectrogram, precomputed once and stored as float16:

$$S = \mathrm{dB}_{80}\Big(\mathrm{Mel}_{256}\big(|\mathrm{STFT}_{2048,512}(x)|^2\big)\Big), \qquad 20 \le f \le 16{,}000 \text{ Hz}, \tag{1}$$

with $n_{\mathrm{fft}} = 2048$, hop 512, 256 mel bands and an 80 dB dynamic-range floor. The result is $256 \times 313$ per window. The default model input is a 3-channel stack $[S,\ \Delta S,\ \Delta^2 S]$, where $\Delta$ is torchaudio's regression-based temporal derivative. Each channel is standardised per sample to zero mean and unit variance over (frequency, time). Figure 3 illustrates the three channels. Stacking three channels lets the ImageNet `conv1` weights be used unchanged. The `single_channel_mel` ablation uses $S$ alone, with `conv1` initialised as the RGB-mean filter.

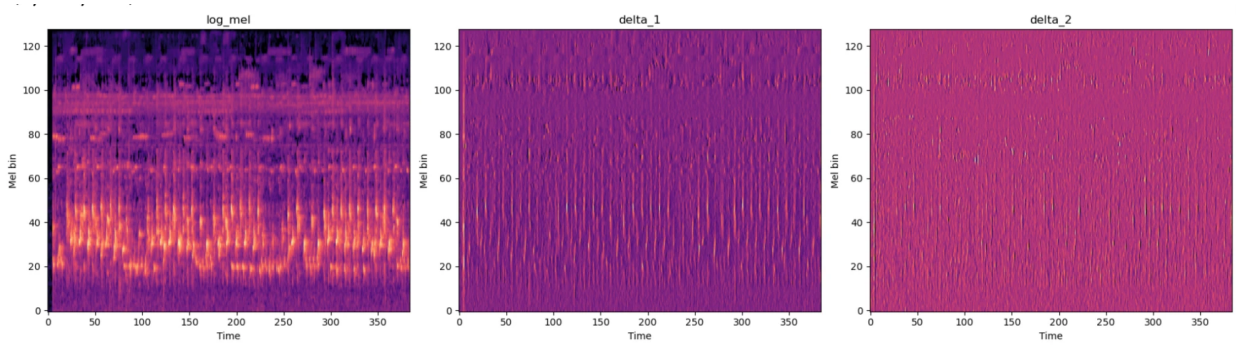



Figure 3: Illustration of the three input channels: log-mel (left), first-order delta (centre) and second-order delta (right). This example was rendered at an earlier, coarser resolution (128 mel bins); the study uses 256 mel bins and 313 frames.

## 3.2 Student network

**Backbone.** An ImageNet-pretrained ResNet-18 [4] truncated after `layer4`, with the global average pool and fully connected layer removed. For a $256 \times 313$ input it outputs a feature map $H \in \mathbb{R}^{512\times 8\times 10}$ (channels $\times$ frequency $\times$ time). Each time step covers about 0.5 s.

**SED head (default, `HEAD=sed`).**

1. **GeM frequency pooling** with a learnable exponent $p$ (initialised at 3, clamped at $p \ge 1$): $h_t = \big(\frac{1}{F}\sum_f H_{:,f,t}^{p}\big)^{1/p}$. With $p = 1$ fixed this becomes mean pooling (`mean_freq_pool`).
2. A bottleneck: Dropout(0.25) $\to$ Linear(512$\to$512) $\to$ ReLU $\to$ Dropout(0.5), applied at each time step.
3. Two $1 \times 1$ temporal convolutions give attention logits $a_{c,t}$ and frame logits $z_{c,t}$ for each class $c$:

$$\alpha_{c,t} = \frac{\exp(\tanh a_{c,t})}{\sum_{t'} \exp(\tanh a_{c,t'})}, \qquad \hat{y}_c^{\mathrm{clip}} = \sum_t \alpha_{c,t}\, z_{c,t}, \qquad \hat{y}_c^{\mathrm{fmax}} = \max_t z_{c,t}. \tag{2}$$

`mean_time_pool` replaces $\alpha_{c,t}$ with $1/T$.

At inference the window probability is the blend $p_c = \frac{1}{2}\sigma(\hat{y}_c^{\mathrm{clip}}) + \frac{1}{2}\sigma(\hat{y}_c^{\mathrm{fmax}})$. The post-hoc variants `head_clip_only` and `head_framemax_only` use one term each.

**Pooled baseline (`HEAD=pooled`).** The plain ResNet-18: frequency-averaged features, dropout(0.2) and a linear classifier per time step, with clip logits taken as the mean over time. No distillation is used.

**Distillation head.** A training-only branch, GAP($H$) →Linear(512→1536), regresses the cached Perch v2 embedding $e^{\text{Perch}}$ of the same window [3, 6]:

$$\mathcal{L}_{\text{dist}} = \frac{1}{1536}\|W\,\text{GAP}(H) - e^{\text{Perch}}\|_2^2. \tag{3}$$

**Stop-gradient.** When distillation is on (default), the classification head reads stopgrad($H$). *The backbone is therefore trained only by the distillation loss*, and the classification loss updates only the head. `distill_no_stopgrad` lets both losses update the backbone; `no_distill` removes the branch, so the backbone trains end-to-end on classification. The branch is dropped at export, so distillation adds nothing to inference cost.

### 3.3 Perch pseudo-labels

Perch v2 is run once, offline and on the GPU, over every 5 s window. It supplies a 1536-d embedding (the distillation target), per-frame embeddings ($16 \times 1536$, for the probe baseline) and class logits. Perch classes are mapped to competition classes by scientific name. Unmapped non-sonotype species in Aves, Amphibia and Insecta use a *genus proxy*: the maximum over Perch classes of the same genus.

**Focal windows (`USE_PSEUDO_FOCAL`).** A focal clip's labels say *which* species are present, and Perch says *when*. For each labelled species $c$ of a clip with windows $w$,

$$\tilde{y}_{w,c} = \frac{p_c(w)}{\max_{w'} p_c(w')}, \tag{4}$$

and every other class is set to 0. If Perch cannot detect the species anywhere in the clip ($\max_w p_c(w) < 0.05$), every window keeps the hard label 1. The ablation `no_pseudo_focal` uses the clip's hard primary + secondary labels for every window.

**Unlabelled soundscapes (`USE_PSEUDO_SC`).** Perch scores are fused in logit space with a shrunk site/hour prior (described next). The weights are $\lambda = 0.4$ for mapped "event" taxa (birds), 1.0 for mapped "texture" taxa (frogs, insects) and 0.8 for texture genus proxies. Classes with no Perch mapping receive the prior alone, and unmapped classes never seen in labelled soundscapes are set to $-8$. Scores are then smoothed across the 12 windows of a file (neighbour averaging 0.35 for texture taxa, local-max 0.15 for event taxa) and passed through a sigmoid to give soft targets. The prior tables for fold $k$ exclude fold-$k$ files.

**Site/hour prior.** From labelled windows, species frequencies are tabulated globally, per hour, per site and per (site, hour). They are shrunk hierarchically as global → hour ($k = 8$) → site ($k = 8$) → site-hour ($k = 4$), with the weight of each level equal to $n/(n + k)$, and converted to logits.

### 3.4 Training

- **Data streams and batch composition.** Each batch of 64 is drawn with fixed shares: 70% focal windows, 10% labelled soundscape windows (other folds) and 20% pseudo-labelled soundscape windows, renormalised when a stream is disabled. For fold 0 the streams hold 121,805 focal, 624 labelled-soundscape and 24,000 pseudo windows; folds 1 and 2 are similar.
- **Rare upsampling.** Species with fewer than 20 focal windows are repeated up to 20. This affects 15 species and grows the focal windows from 152,198 to 152,431. `no_rare_upsample` sets the minimum to 0.
- **Mixup** [7]. Focal–focal mixup ($p = 0.5$, $\lambda \sim \mathrm{Beta}(0.4, 0.4)$) and focal–soundscape mixup with a labelled soundscape window from another fold ($p = 0.5$, $\mathrm{Beta}(0.4, 0.4)$). Mixing is done in the power domain, $10 \log_{10}(\lambda 10^{a/10} + (1-\lambda)10^{b/10})$, which is equivalent to mixing the audio. Labels are combined by union ("hard" mixup), and Perch embeddings are mixed with the same $\lambda$.
- **SpecAugment** [5]: one frequency mask ($\leq$10 bins) and two time masks ($\leq$10 frames).
- **Loss.** $\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + 1.0 \cdot \mathcal{L}_{\mathrm{dist}}$, where $\mathcal{L}_{\mathrm{cls}} = \frac{1}{2}\mathrm{BCE}(\hat{y}^{\mathrm{clip}}, \tilde{y}) + \frac{1}{2}\mathrm{BCE}(\hat{y}^{\mathrm{fmax}}, \tilde{y})$, averaged over classes and weighted per source (focal, labelled soundscape and pseudo soundscape all 1.0).
- **Optimisation.** AdamW, constant learning rate $5 \times 10^{-4}$, weight decay $10^{-4}$, gradient clipping at norm 1.0, mixed precision, seed 42.
- **Budget.** 10 epochs of 600 batches (6,000 steps, about 2.6 passes over the focal windows), cut from the notebook's 25 full epochs to make the ablations affordable.
- **Checkpoint selection.** After every epoch the model is scored on its validation fold, and the epoch with the best macro ROC-AUC over *all* that fold's soundscape windows (S22 included) is kept.

The waveform gain and noise augmentation in the original notebook were dropped when spectrograms were precomputed.

### 3.5 Baselines

`resnet_baseline` Pooled ResNet-18 head, no distillation; all other settings as in `full`.

`perch_attn_probe` Frozen Perch v2 per-frame embeddings ($16 \times 1536$) $\rightarrow$ LayerNorm $\rightarrow$ the same attention-pooling head without the hidden layer. It uses the same training streams, targets and folds and is trained for 30 epochs (batch 256, AdamW, learning rate $10^{-3}$, weight decay $10^{-4}$). It answers how good the teacher alone is, and what running Perch on site would cost.

### 3.6 Study design

Each training variant changes exactly one setting relative to `full` (Table 1). Post-hoc variants reuse `full`'s out-of-fold predictions.

Table 1: The 20 variants.

| Variant | Change relative to `full` |
|---|---|
| `resnet_baseline` | Pooled ResNet-18 head, no distillation |
| `perch_attn_probe` | Frozen Perch v2 + attention-pooling linear probe (no CNN) |
| `full` | All components on |
| `no_distill` | No Perch distillation (backbone trained by classification) |
| `distill_no_stopgrad` | Distillation on, but classification gradients also reach the backbone |
| `no_pseudo_focal` | Hard clip labels on every focal window |
| `no_pseudo_sc` | No pseudo-labelled soundscape stream |
| `single_channel_mel` | Log-mel only (1 channel) |
| `mean_freq_pool` | Mean instead of GeM over frequency |
| `mean_time_pool` | Uniform instead of attention weights over time |
| `no_mixup` | No focal–focal or focal–soundscape mixup |
| `no_soundscape_data` | No labelled soundscapes, no focal–soundscape mixup, no pseudo soundscapes |
| `no_specaug` | No frequency/time masking |
| `no_rare_upsample` | No upsampling of species with <20 windows |
| `head_clip_only` | Attention-pooled clip probability only |
| `head_framemax_only` | Frame-max probability only |
| `full+gauss_smooth` | 5-tap kernel $(0.1, 0.2, 0.4, 0.2, 0.1)$ over neighbouring windows, in logit space |
| `full+tax_smooth` | Pull each species toward its genus mean ($\alpha = 0.15$), then class mean ($\alpha = 0.05$) |
| `full+site_hour_prior` | Add fold-safe site/hour prior logits ($\lambda = 1.0$ frogs/insects, 0.4 others) |
| `full+all_posthoc` | Prior → Gaussian smoothing → taxonomy smoothing |

### 3.7 Metrics

All metrics are computed on the pooled out-of-fold predictions for the 192 evaluation windows (Section 2.4), ordered by file and time. Classes without both a positive and a negative are skipped.

- **Macro ROC-AUC** (primary): mean of per-class ROC-AUC.
- **Macro PR-AUC**: mean per-class average precision. The chance level equals class prevalence (0.069 on average here).
- **LRAP**: label-ranking average precision, i.e. how well the true species are ranked within each window.
- **Macro F1** at a threshold of 0.5.
- **ECE**: expected calibration error with 15 bins over all (window, class) cells. **Brier**: mean squared error of the probabilities.
- **Temporal jitter**: mean $|p_t - p_{t-1}|$ between consecutive windows of the same file.
- **AUC rare / common**: macro ROC-AUC over classes with <20 / ≥100 focal clips.
- **AUC per taxon**: macro ROC-AUC over Aves, Amphibia, Insecta and Mammalia classes. Reptilia has no scorable class here.

### 3.8 Cost measurement and hardware

Training ran on one NVIDIA L4 (GCP `g2-standard-16`) with three runs sharing the GPU at a time, each with 4 data-loader workers. The offline Perch pass took 830 s of wall time (0.231 GPU-hours) for the 24,792 soundscape windows. Inference cost (`benchmark_cost.py`) rebuilds each network from its `config.json` and times features + network on random input: single-thread and 4-thread CPU at batch 12 (one 60 s file), and GPU at batch 64. Perch is timed on the GPU only. The trained fold checkpoints were exported to ONNX and checked against PyTorch. All 30 exports recorded in the logs agree to a maximum relative logit difference of $2.2 \times 10^{-4}$.

## 4 Results

### 4.1 Main results

Tables 2 and 3 give every metric for every variant. Figure 4 summarises the primary metric.

Table 2: Ranking and detection metrics for every variant on the pooled out-of-fold non-S22 soundscape windows (higher is better; best per column in bold).

| Variant | Macro ROC-AUC | Macro PR-AUC | LRAP | Macro F1 | AUC rare | AUC common |
|---|---|---|---|---|---|---|
| *Baselines* | | | | | | |
| `resnet_baseline` | 0.7571 | 0.3618 | 0.2772 | **0.2522** | 0.7104 | 0.8387 |
| `perch_attn_probe` | 0.7109 | 0.3991 | 0.1390 | 0.1654 | 0.5778 | **0.9042** |
| *Full model* | | | | | | |
| `full` | 0.7142 | 0.3634 | 0.2209 | 0.1337 | 0.6679 | 0.7654 |
| *One-change training ablations* | | | | | | |
| `no_distill` | 0.7292 | 0.3308 | 0.1820 | 0.1013 | 0.7004 | 0.7697 |
| `distill_no_stopgrad` | 0.6970 | 0.3738 | 0.2238 | 0.2349 | 0.6823 | 0.6460 |
| `no_pseudo_focal` | 0.7049 | 0.3675 | 0.1887 | 0.1261 | 0.6876 | 0.7004 |
| `no_pseudo_sc` | 0.7742 | **0.4321** | **0.4770** | 0.2396 | 0.7417 | 0.8174 |
| `single_channel_mel` | 0.7230 | 0.3744 | 0.1862 | 0.1446 | 0.6841 | 0.7714 |
| `mean_freq_pool` | 0.7466 | 0.4048 | 0.2897 | 0.2178 | 0.7165 | 0.7825 |
| `mean_time_pool` | 0.6984 | 0.3687 | 0.2101 | 0.1260 | 0.6651 | 0.7277 |
| `no_mixup` | 0.6630 | 0.3396 | 0.1596 | 0.0866 | 0.6270 | 0.6813 |
| `no_soundscape_data` | 0.6656 | 0.1874 | 0.0589 | 0.0000 | 0.5822 | 0.8675 |
| `no_specaug` | 0.7135 | 0.3607 | 0.2789 | 0.1692 | 0.6685 | 0.7538 |
| `no_rare_upsample` | 0.6896 | 0.3719 | 0.1956 | 0.1331 | 0.6636 | 0.6784 |
| *Post-hoc variants of `full` (no retraining)* | | | | | | |
| `head_clip_only` | 0.7050 | 0.3546 | 0.2267 | 0.0917 | 0.6703 | 0.7239 |
| `head_framemax_only` | 0.7145 | 0.3642 | 0.2141 | 0.1363 | 0.6620 | 0.7810 |
| `full+gauss_smooth` | 0.7206 | 0.3940 | 0.2185 | 0.1221 | 0.6772 | 0.7622 |
| `full+tax_smooth` | 0.7632 | 0.3743 | 0.2092 | 0.1046 | 0.7308 | 0.7631 |
| `full+site_hour_prior` | 0.7346 | 0.3608 | 0.3964 | 0.1575 | 0.6962 | 0.7432 |
| `full+all_posthoc` | **0.8407** | 0.4159 | 0.4082 | 0.1164 | **0.8335** | 0.7987 |

Table 3: Calibration, temporal stability and per-taxon macro ROC-AUC for every variant (ECE, Brier and jitter: lower is better). Best per column in bold.

| Variant | ECE↓ | Brier↓ | Jitter↓ | AUC Aves | AUC Amph. | AUC Insecta | AUC Mamm. |
|---|---|---|---|---|---|---|---|
| *Baselines* | | | | | | | |
| `resnet_baseline` | 0.2256 | 0.1503 | 0.0684 | 0.8064 | 0.8347 | 0.6652 | 0.8245 |
| `perch_attn_probe` | 0.3564 | 0.2216 | 0.0860 | **0.8972** | 0.8625 | 0.4908 | 0.8185 |
| *Full model* | | | | | | | |
| `full` | 0.1836 | 0.0931 | 0.0437 | 0.7509 | 0.8442 | 0.6401 | 0.4516 |
| *One-change training ablations* | | | | | | | |
| `no_distill` | 0.2249 | 0.1462 | 0.0620 | 0.7651 | 0.7791 | 0.7045 | 0.4322 |
| `distill_no_stopgrad` | 0.2501 | 0.1691 | 0.0519 | 0.6709 | 0.7802 | 0.6797 | 0.4890 |
| `no_pseudo_focal` | 0.2136 | 0.1152 | 0.0407 | 0.6849 | 0.8227 | 0.6512 | 0.6135 |
| `no_pseudo_sc` | **0.0032** | **0.0092** | 0.0027 | 0.8026 | 0.8961 | 0.6673 | **0.8447** |
| `single_channel_mel` | 0.2509 | 0.1467 | 0.0397 | 0.7572 | 0.8340 | 0.6382 | 0.6638 |
| `mean_freq_pool` | 0.1808 | 0.0931 | 0.0454 | 0.7709 | 0.8568 | 0.6548 | 0.7771 |
| `mean_time_pool` | 0.2183 | 0.1131 | 0.0391 | 0.7201 | 0.7993 | 0.6394 | 0.4956 |
| `no_mixup` | 0.2377 | 0.1246 | 0.0494 | 0.6614 | 0.8192 | 0.6035 | 0.4272 |
| `no_soundscape_data` | 0.0078 | 0.0128 | 0.0016 | 0.8078 | 0.5871 | 0.6478 | 0.4066 |
| `no_specaug` | 0.2180 | 0.1204 | 0.0403 | 0.7616 | 0.8277 | 0.6090 | 0.6939 |
| `no_rare_upsample` | 0.2281 | 0.1245 | 0.0441 | 0.7038 | 0.8163 | 0.6136 | 0.5735 |
| *Post-hoc variants of `full` (no retraining)* | | | | | | | |
| `head_clip_only` | 0.1477 | 0.0733 | 0.0404 | 0.7171 | 0.8367 | 0.6430 | 0.4525 |
| `head_framemax_only` | 0.2196 | 0.1176 | 0.0505 | 0.7638 | 0.8457 | 0.6349 | 0.4364 |
| `full+gauss_smooth` | 0.1814 | 0.0902 | 0.0124 | 0.7468 | 0.8573 | 0.6485 | 0.4668 |
| `full+tax_smooth` | 0.1836 | 0.0917 | 0.0434 | 0.7450 | **0.9173** | 0.7217 | 0.4287 |
| `full+site_hour_prior` | 0.0059 | 0.0102 | 0.0045 | 0.7591 | 0.8383 | 0.6599 | 0.6240 |
| `full+all_posthoc` | 0.0062 | 0.0100 | **0.0013** | 0.7943 | 0.9139 | **0.8558** | 0.5662 |

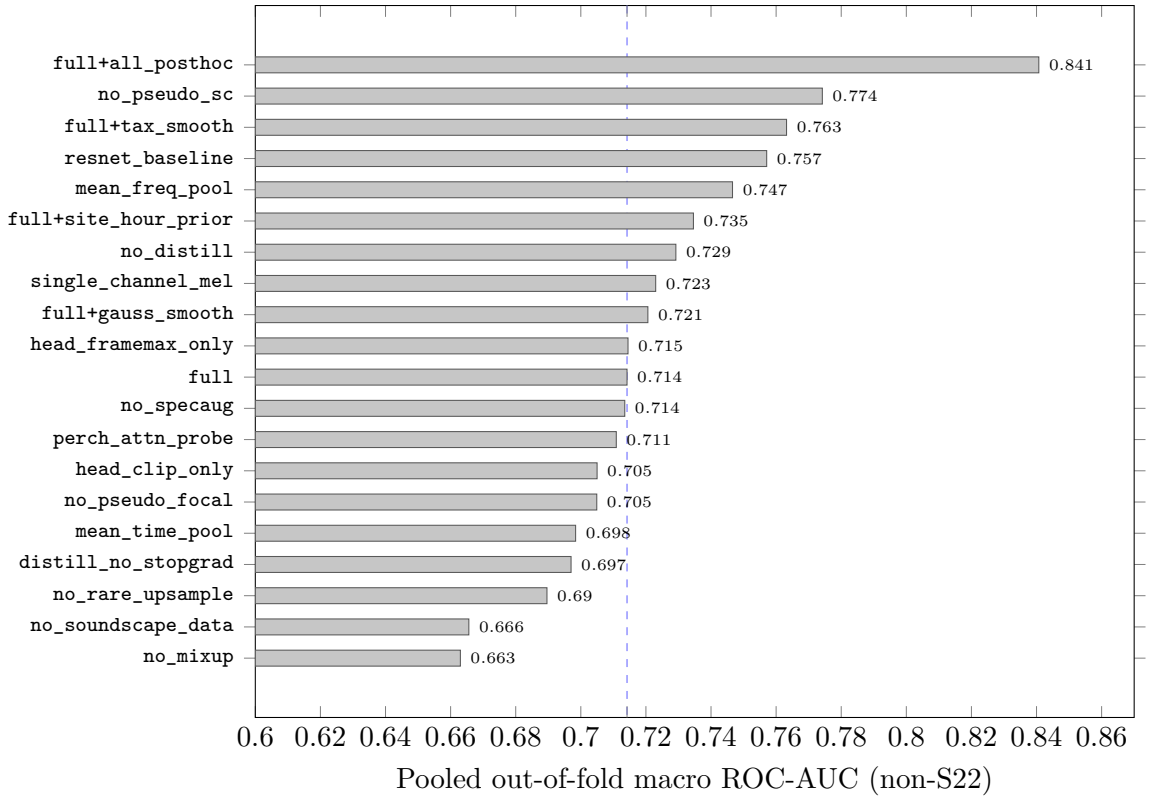



Figure 4: All 20 variants ranked by the primary metric. The dashed line marks `full` (0.7142).

The headline results are:

- **Best overall:** `full+all_posthoc` has a macro ROC-AUC of 0.8407, rare-class AUC 0.8335, Insecta 0.8558, ECE 0.0062 and jitter 0.0013, at the inference cost of `full`.
- **Best trained model:** `no_pseudo_sc` leads on macro AUC (0.7742), PR-AUC (0.4321), LRAP (0.4770), ECE (0.0032), Brier (0.0092) and Mammalia AUC (0.8447).
- **The plain baseline beats the full model:** `resnet_baseline` reaches 0.7571 (+0.0429 over `full`) and has the best macro F1 (0.2522).
- **The teacher alone** (`perch_attn_probe`) is best on common species (0.9042) and birds (0.8972), but worst on rare species (0.5778), insects (0.4908), ECE (0.3564), Brier (0.2216) and jitter (0.0860), and second-worst on LRAP (0.1390).
- **The two most damaging removals** are mixup (0.6630) and soundscape data (0.6656). The latter predicts nothing above 0.5 (F1 = 0.0000) and has PR-AUC 0.1874, but has the second-highest common-species AUC (0.8675).

## 4.2 Changes relative to the full model

Tables 4 and 5 give the change from `full` for every metric.

Table 4: Change relative to `full` (variant − full) for the ranking metrics.

| Variant | ΔAUC | ΔPR-AUC | ΔLRAP | ΔF1 | Δrare | Δcommon |
|---|---|---|---|---|---|---|
| *Baselines* | | | | | | |
| `resnet_baseline` | +0.0429 | −0.0016 | +0.0563 | +0.1185 | +0.0425 | +0.0733 |
| `perch_attn_probe` | −0.0032 | +0.0356 | −0.0820 | +0.0317 | −0.0901 | +0.1388 |
| *One-change training ablations* | | | | | | |
| `no_distill` | +0.0150 | −0.0327 | −0.0389 | −0.0323 | +0.0325 | +0.0044 |
| `distill_no_stopgrad` | −0.0172 | +0.0103 | +0.0028 | +0.1013 | +0.0144 | −0.1194 |
| `no_pseudo_focal` | −0.0093 | +0.0040 | −0.0323 | −0.0076 | +0.0196 | −0.0650 |
| `no_pseudo_sc` | +0.0600 | +0.0686 | +0.2560 | +0.1059 | +0.0738 | +0.0521 |
| `single_channel_mel` | +0.0088 | +0.0110 | −0.0347 | +0.0109 | +0.0162 | +0.0060 |
| `mean_freq_pool` | +0.0325 | +0.0414 | +0.0688 | +0.0841 | +0.0486 | +0.0171 |
| `mean_time_pool` | −0.0158 | +0.0053 | −0.0108 | −0.0077 | −0.0028 | −0.0377 |
| `no_mixup` | −0.0512 | −0.0239 | −0.0614 | −0.0471 | −0.0409 | −0.0841 |
| `no_soundscape_data` | −0.0486 | −0.1760 | −0.1620 | −0.1337 | −0.0857 | +0.1021 |
| `no_specaug` | −0.0007 | −0.0027 | +0.0579 | +0.0355 | +0.0006 | −0.0116 |
| `no_rare_upsample` | −0.0245 | +0.0085 | −0.0254 | −0.0006 | −0.0043 | −0.0870 |
| *Post-hoc variants of `full` (no retraining)* | | | | | | |
| `head_clip_only` | −0.0092 | −0.0089 | +0.0058 | −0.0420 | +0.0024 | −0.0415 |
| `head_framemax_only` | +0.0003 | +0.0008 | −0.0068 | +0.0026 | −0.0059 | +0.0157 |
| `full+gauss_smooth` | +0.0064 | +0.0306 | −0.0024 | −0.0116 | +0.0093 | −0.0032 |
| `full+tax_smooth` | +0.0490 | +0.0108 | −0.0117 | −0.0291 | +0.0629 | −0.0023 |
| `full+site_hour_prior` | +0.0204 | −0.0027 | +0.1754 | +0.0238 | +0.0282 | −0.0221 |
| `full+all_posthoc` | +0.1266 | +0.0524 | +0.1873 | −0.0173 | +0.1656 | +0.0333 |

Table 5: Change relative to `full` for calibration, jitter and per-taxon AUC (for ECE, Brier and jitter a negative change is an improvement).

| Variant | ΔECE | ΔBrier | ΔJitter | ΔAves | ΔAmph. | ΔInsecta | ΔMamm. |
|---|---|---|---|---|---|---|---|
| *Baselines* | | | | | | | |
| `resnet_baseline` | +0.0419 | +0.0573 | +0.0247 | +0.0556 | −0.0095 | +0.0251 | +0.3729 |
| `perch_attn_probe` | +0.1727 | +0.1285 | +0.0423 | +0.1463 | +0.0183 | −0.1493 | +0.3670 |
| *One-change training ablations* | | | | | | | |
| `no_distill` | +0.0413 | +0.0531 | +0.0183 | +0.0142 | −0.0651 | +0.0643 | −0.0194 |
| `distill_no_stopgrad` | +0.0665 | +0.0760 | +0.0082 | −0.0800 | −0.0640 | +0.0396 | +0.0374 |
| `no_pseudo_focal` | +0.0299 | +0.0221 | −0.0030 | −0.0660 | −0.0215 | +0.0111 | +0.1619 |
| `no_pseudo_sc` | −0.1804 | −0.0839 | −0.0411 | +0.0517 | +0.0519 | +0.0272 | +0.3931 |
| `single_channel_mel` | +0.0673 | +0.0536 | −0.0040 | +0.0063 | −0.0102 | −0.0019 | +0.2122 |
| `mean_freq_pool` | −0.0028 | −0.0000 | +0.0017 | +0.0201 | +0.0126 | +0.0147 | +0.3255 |
| `mean_time_pool` | +0.0347 | +0.0200 | −0.0046 | −0.0308 | −0.0450 | −0.0007 | +0.0440 |
| `no_mixup` | +0.0540 | +0.0315 | +0.0057 | −0.0894 | −0.0250 | −0.0366 | −0.0244 |
| `no_soundscape_data` | −0.1759 | −0.0803 | −0.0421 | +0.0569 | −0.2571 | +0.0077 | −0.0450 |
| `no_specaug` | +0.0343 | +0.0273 | −0.0034 | +0.0108 | −0.0165 | −0.0312 | +0.2423 |
| `no_rare_upsample` | +0.0445 | +0.0314 | +0.0004 | −0.0470 | −0.0279 | −0.0265 | +0.1219 |
| *Post-hoc variants of `full` (no retraining)* | | | | | | | |
| `head_clip_only` | −0.0360 | −0.0198 | −0.0033 | −0.0338 | −0.0075 | +0.0029 | +0.0009 |
| `head_framemax_only` | +0.0360 | +0.0245 | +0.0068 | +0.0129 | +0.0015 | −0.0052 | −0.0152 |
| `full+gauss_smooth` | −0.0023 | −0.0029 | −0.0313 | −0.0041 | +0.0130 | +0.0084 | +0.0152 |
| `full+tax_smooth` | −0.0000 | −0.0014 | −0.0003 | −0.0059 | +0.0731 | +0.0816 | −0.0229 |
| `full+site_hour_prior` | −0.1777 | −0.0829 | −0.0392 | +0.0082 | −0.0059 | +0.0198 | +0.1724 |
| `full+all_posthoc` | −0.1774 | −0.0831 | −0.0424 | +0.0435 | +0.0697 | +0.2157 | +0.1146 |

### 4.3 Statistical uncertainty

The evaluation set has only 16 files, and windows within a file are strongly correlated. We therefore ran a *file-level* bootstrap: the 16 files were resampled with replacement 2,000 times (seed 0), and macro ROC-AUC was recomputed for every trained variant on each resample. Paired differences against `full` use the same resamples. The post-hoc variants need the competition taxonomy file, which is not stored with the results, so they are not bootstrapped.

Table 6: File-level bootstrap (2,000 resamples of the 16 evaluation files) of pooled macro ROC-AUC. The 95% interval is the 2.5–97.5 percentile range. $\Delta$ is the paired difference to `full` on the same resamples; $P(\Delta > 0)$ is the fraction of resamples in which the variant beats `full`. Intervals whose $\Delta$ range excludes 0 are marked $*$.

| Variant | Macro AUC | 95% CI | $\Delta$ vs full | 95% CI of $\Delta$ | $P(\Delta > 0)$ |
|---|---|---|---|---|---|
| `no_pseudo_sc` | 0.7742 | [0.703, 0.872] | +0.0600 | [−0.0009, +0.1513] | 0.974 |
| `resnet_baseline` | 0.7571 | [0.665, 0.858] | +0.0429 | [−0.0114, +0.1098] | 0.943 |
| `mean_freq_pool` | 0.7466 | [0.662, 0.849] | +0.0325 | [+0.0018, +0.0735]$*$ | 0.979 |
| `no_distill` | 0.7292 | [0.632, 0.824] | +0.0150 | [−0.0408, +0.0794] | 0.675 |
| `single_channel_mel` | 0.7230 | [0.637, 0.826] | +0.0088 | [−0.0281, +0.0527] | 0.690 |
| `full` | 0.7142 | [0.622, 0.820] | – | – | – |
| `no_specaug` | 0.7135 | [0.626, 0.808] | −0.0007 | [−0.0413, +0.0456] | 0.468 |
| `perch_attn_probe` | 0.7109 | [0.622, 0.798] | −0.0032 | [−0.0945, +0.0737] | 0.408 |
| `no_pseudo_focal` | 0.7049 | [0.604, 0.814] | −0.0093 | [−0.0481, +0.0281] | 0.274 |
| `mean_time_pool` | 0.6984 | [0.606, 0.811] | −0.0158 | [−0.0414, +0.0196] | 0.230 |
| `distill_no_stopgrad` | 0.6970 | [0.580, 0.798] | −0.0172 | [−0.0886, +0.0389] | 0.249 |
| `no_rare_upsample` | 0.6896 | [0.600, 0.793] | −0.0245 | [−0.0573, +0.0213] | 0.133 |
| `no_soundscape_data` | 0.6656 | [0.582, 0.723] | −0.0486 | [−0.1919, +0.0503] | 0.151 |
| `no_mixup` | 0.6630 | [0.559, 0.764] | −0.0512 | [−0.1113, −0.0144]$*$ | 0.006 |

The intervals are wide. `full`'s macro AUC has a 95% interval of [0.622, 0.820], about $\pm 0.10$. Only two paired differences exclude zero: `mean_freq_pool` is better than `full` (+0.0325, interval [+0.002, +0.074], better in 97.9% of resamples) and `no_mixup` is worse (−0.0512, interval [−0.111, −0.014], better in only 0.6% of resamples). `no_pseudo_sc` (+0.0600, better in 97.4% of resamples, interval [−0.001, +0.151]) and `resnet_baseline` (+0.0429, 94.3%) are borderline. Every other difference in macro AUC, including `no_soundscape_data` (−0.0486, interval [−0.192, +0.050]), is within resampling noise for this metric.

### 4.4 Pooled versus fold-local scores

Each fold's checkpoint is chosen on that fold's validation windows (Table 7). Scored *within* a fold, the non-S22 macro AUC of the selected checkpoints is 0.70–0.94 (0.54 for `no_soundscape_data` on fold 1). For `full` the mean of the three folds is 0.825. The pooled out-of-fold scores in Table 2 are much lower (0.66–0.77). Table 8 gives the fold-local macro AUC on the evaluation windows.

Table 7: Per-fold checkpoint selection for the CNN variants (`history.json`). The saved checkpoint is the epoch with the best macro ROC-AUC over *all* validation windows of that fold (S22 included); "non-S22 @best" is the fold-local non-S22 macro AUC of that checkpoint and "max non-S22" the best value reached in any epoch. Fold-local numbers are higher than the pooled out-of-fold score (Section 4.4).

| | Fold 0 | | | Fold 1 | | | Fold 2 | | |
|---|---|---|---|---|---|---|---|---|---|
| Variant | best ep | non-S22 @best | max non-S22 | best ep | non-S22 @best | max non-S22 | best ep | non-S22 @best | max non-S22 |
| `resnet_baseline` | 3 | 0.8808 | 0.8808 | 9 | 0.8606 | 0.9073 | 9 | 0.8781 | 0.8781 |
| `full` | 5 | 0.8177 | 0.8214 | 9 | 0.8385 | 0.8499 | 6 | 0.8196 | 0.8476 |
| `no_distill` | 2 | 0.8233 | 0.8491 | 7 | 0.8950 | 0.8950 | 5 | 0.8461 | 0.8461 |
| `distill_no_stopgrad` | 9 | 0.7228 | 0.8030 | 6 | 0.8508 | 0.8545 | 9 | 0.7943 | 0.7943 |
| `no_pseudo_focal` | 5 | 0.8004 | 0.8004 | 8 | 0.8645 | 0.8645 | 8 | 0.8483 | 0.8483 |
| `no_pseudo_sc` | 9 | 0.9228 | 0.9228 | 6 | 0.8955 | 0.8976 | 9 | 0.9412 | 0.9412 |
| `single_channel_mel` | 8 | 0.8852 | 0.8959 | 8 | 0.8802 | 0.8862 | 8 | 0.8475 | 0.8696 |
| `mean_freq_pool` | 9 | 0.8838 | 0.8838 | 9 | 0.8780 | 0.8955 | 9 | 0.8777 | 0.8844 |
| `mean_time_pool` | 8 | 0.8021 | 0.8315 | 9 | 0.8564 | 0.8790 | 9 | 0.8780 | 0.8780 |
| `no_mixup` | 8 | 0.7384 | 0.7557 | 8 | 0.8210 | 0.8210 | 9 | 0.7998 | 0.7998 |
| `no_soundscape_data` | 9 | 0.7436 | 0.7436 | 1 | 0.5387 | 0.6713 | 9 | 0.7241 | 0.7241 |
| `no_specaug` | 8 | 0.8016 | 0.8016 | 9 | 0.8852 | 0.8852 | 8 | 0.8582 | 0.8748 |
| `no_rare_upsample` | 8 | 0.7902 | 0.7902 | 7 | 0.8816 | 0.8816 | 9 | 0.8567 | 0.8567 |

Table 8: Fold-local macro ROC-AUC: each fold's selected checkpoint scored only on that fold's non-S22 evaluation windows, versus the pooled out-of-fold score over all 192 windows.

| Variant | Fold 0 | Fold 1 | Fold 2 | Mean of folds | Pooled |
|---|---|---|---|---|---|
| `no_pseudo_sc` | 0.9228 | 0.8955 | 0.9412 | 0.9198 | 0.7742 |
| `resnet_baseline` | 0.8808 | 0.8606 | 0.8781 | 0.8732 | 0.7571 |
| `mean_freq_pool` | 0.8838 | 0.8780 | 0.8777 | 0.8799 | 0.7466 |
| `no_distill` | 0.8233 | 0.8950 | 0.8461 | 0.8548 | 0.7292 |
| `single_channel_mel` | 0.8852 | 0.8802 | 0.8475 | 0.8710 | 0.7230 |
| `full` | 0.8177 | 0.8385 | 0.8196 | 0.8253 | 0.7142 |
| `no_specaug` | 0.8016 | 0.8852 | 0.8582 | 0.8483 | 0.7135 |
| `perch_attn_probe` | 0.7495 | 0.7015 | 0.8207 | 0.7572 | 0.7109 |
| `no_pseudo_focal` | 0.8004 | 0.8645 | 0.8483 | 0.8378 | 0.7049 |
| `mean_time_pool` | 0.8021 | 0.8564 | 0.8780 | 0.8455 | 0.6984 |
| `distill_no_stopgrad` | 0.7228 | 0.8508 | 0.7943 | 0.7893 | 0.6970 |
| `no_rare_upsample` | 0.7902 | 0.8816 | 0.8567 | 0.8428 | 0.6896 |
| `no_soundscape_data` | 0.7436 | 0.5387 | 0.7241 | 0.6688 | 0.6656 |
| `no_mixup` | 0.7384 | 0.8210 | 0.7998 | 0.7864 | 0.6630 |

### 4.5 Pre-registered hypotheses

Before training, each variant was given the metric it should move most and the expected direction (Table 9). **13 of 21** hypotheses were supported.

Table 9: Pre-registered hypotheses (one per row, from `configs.py`) against the observed change relative to `full`. "Biggest-Δ" is the metric that moved most in absolute terms. The rationale written before the runs is given in italics.

| Variant | Metric | Predicted | Δ vs full | Biggest-Δ | Verdict |
|---|---|---|---|---|---|
| `resnet_baseline` | `lrap` | drop | +0.0563 | `auc_Mammalia` | not supported |
| *A plain pooled ResNet loses the time axis, so it ranks species within a window worse (LRAP) and is less calibrated than the attention SED head at the same cost.* | | | | | |
| `resnet_baseline` | `ece` | rise | +0.0419 | `auc_Mammalia` | **supported** |
| *Without distillation or frame-level heads the baseline overfits the small label set.* | | | | | |
| `perch_attn_probe` | `auc_rare` | rise | −0.0901 | `auc_Mammalia` | not supported |
| *Perch has seen far more data; the frozen teacher should beat the student on rare species, at many times the inference cost (see benchmark_cost.py).* | | | | | |
| `no_pseudo_focal` | `pr_auc_macro` | drop | +0.0040 | `auc_Mammalia` | not supported |
| *Clip-level labels mark every 5 s window as positive even when the bird is silent; Perch-localised targets remove those false positives, mainly helping precision (PR-AUC).* | | | | | |
| `no_pseudo_sc` | `auc_soundscape_macro` | drop | +0.0600 | `auc_Mammalia` | not supported |
| *Unlabelled soundscapes are in-domain; their pseudo-labels should help soundscape AUC and calibration most.* | | | | | |
| `single_channel_mel` | `auc_Insecta` | drop | −0.0019 | `auc_Mammalia` | **supported** |
| *Delta channels encode modulation / onsets; insects and frogs are defined by repetition rate, so dropping them should hurt non-bird taxa more.* | | | | | |
| `no_distill` | `auc_rare` | drop | +0.0325 | `auc_Amphibia` | not supported |
| *Perch embeddings carry knowledge about species with few focal clips.* | | | | | |
| `no_distill` | `ece` | rise | +0.0413 | `auc_Amphibia` | **supported** |
| *Without the teacher regulariser the student overfits, giving worse calibration.* | | | | | |
| `distill_no_stopgrad` | `auc_soundscape_macro` | same | −0.0172 | `auc_common` | not supported |
| *Letting the classification loss update the backbone should matter little.* | | | | | |
| `mean_freq_pool` | `auc_Insecta` | drop | +0.0147 | `auc_Mammalia` | not supported |
| *GeM sharpens toward narrow bands; insects/frogs are narrow-band.* | | | | | |
| `mean_time_pool` | `lrap` | drop | −0.0108 | `auc_Amphibia` | **supported** |
| *Attention pooling picks the frames with the call; uniform pooling dilutes short calls.* | | | | | |
| `no_soundscape_data` | `auc_soundscape_macro` | drop | −0.0486 | `auc_Amphibia` | **supported** |
| *Soundscape windows match the deployment domain.* | | | | | |
| `no_mixup` | `f1_macro` | drop | −0.0471 | `auc_Aves` | **supported** |
| *Mixup teaches multi-species windows, so multi-label F1 should drop more than AUC.* | | | | | |
| `no_specaug` | `auc_soundscape_macro` | small drop | −0.0007 | `auc_Mammalia` | **supported** |
| *SpecAugment is a generic regulariser; expect a small uniform drop.* | | | | | |
| `no_rare_upsample` | `auc_rare` | drop | −0.0043 | `auc_Mammalia` | **supported** |
| *Upsampling only touches species with <20 clips.* | | | | | |
| `head_clip_only` | `lrap` | drop | +0.0058 | `f1_macro` | not supported |
| *Clip and frame-max heads make different errors; the 50/50 blend should beat either.* | | | | | |
| `head_framemax_only` | `ece` | rise | +0.0360 | `ece` | **supported** |
| *Max over frames is biased upward, so frame-max-only should be over-confident.* | | | | | |
| `full+gauss_smooth` | `temporal_jitter` | drop | −0.0313 | `temporal_jitter` | **supported** |
| *Smoothing across neighbouring windows mainly reduces prediction jitter.* | | | | | |

| Variant | Metric | Predicted | Δ vs full | Biggest-Δ | Verdict |
|---|---|---|---|---|---|
| `full+tax_smooth` | `auc_rare` | rise | +0.0629 | `auc_Insecta` | **supported** |
| *Rare species borrow evidence from relatives in the same genus.* | | | | | |
| `full+site_hour_prior` | `auc_soundscape_macro` | rise | +0.0204 | `ece` | **supported** |
| *Where and when a recording was made is a strong, free signal for which species can be present – expected to be the best gain per unit of inference cost.* | | | | | |
| `full+all_posthoc` | `auc_soundscape_macro` | rise | +0.1266 | `auc_Insecta` | **supported** |
| *The free post-processing steps stack.* | | | | | |

### 4.6 Inference and training cost

Table 10: Measured inference cost per 5 s window (`benchmark_cost.py`). CPU timings use a batch of 12 windows (one 60 s file) with 1 or 4 threads and include feature extraction ("feat"); GPU timings use a batch of 64 on the NVIDIA L4. Real-time factor = 5000 ms / CPU-1 ms. Perch is GPU-only; for `perch_attn_probe` the parameter count and size are the probe head only, while the GPU time includes running Perch v2.

| Variant | Params (M) | Size (MB) | GFLOPs | CPU1 feat | CPU1 ms | CPU4 ms | GPU ms | GPU preds/s | RT× |
|---|---|---|---|---|---|---|---|---|---|
| `resnet_baseline` | 11.297 | 45.2 | 5.898 | 9.19 | 83.63 | 21.86 | 0.782 | 1279 | 59.8 |
| `perch_attn_probe` | 0.722 | 2.9 | – | – | – | – | 2.569 | 389 | – |
| `full` | 11.679 | 46.7 | 5.906 | 9.02 | 81.92 | 22.02 | 0.785 | 1273 | 61.0 |
| `no_distill` | 11.679 | 46.7 | 5.906 | 8.85 | 81.54 | 21.25 | 0.783 | 1276 | 61.3 |
| `distill_no_stopgrad` | 11.679 | 46.7 | 5.906 | 9.67 | 82.39 | 21.75 | 0.785 | 1275 | 60.7 |
| `no_pseudo_focal` | 11.679 | 46.7 | 5.906 | 9.28 | 82.03 | 22.89 | 0.784 | 1275 | 61.0 |
| `no_pseudo_sc` | 11.679 | 46.7 | 5.906 | 9.17 | 81.99 | 21.35 | 0.784 | 1275 | 61.0 |
| `single_channel_mel` | 11.673 | 46.7 | 5.654 | 8.12 | 80.82 | 22.57 | 0.731 | 1369 | 61.9 |
| `mean_freq_pool` | 11.679 | 46.7 | 5.906 | 9.02 | 81.39 | 22.02 | 0.783 | 1277 | 61.4 |
| `mean_time_pool` | 11.679 | 46.7 | 5.906 | 9.05 | 82.20 | 21.94 | 0.786 | 1273 | 60.8 |
| `no_mixup` | 11.679 | 46.7 | 5.906 | 8.97 | 81.55 | 21.34 | 0.784 | 1275 | 61.3 |
| `no_soundscape_data` | 11.679 | 46.7 | 5.906 | 9.05 | 81.39 | 21.94 | 0.784 | 1275 | 61.4 |
| `no_specaug` | 11.679 | 46.7 | 5.906 | 9.51 | 82.38 | 21.39 | 0.783 | 1277 | 60.7 |
| `no_rare_upsample` | 11.679 | 46.7 | 5.906 | 8.82 | 81.29 | 21.40 | 0.785 | 1274 | 61.5 |

All 12 ResNet-18 variants cost essentially the same: 11.68 M parameters (11.30 M for the pooled baseline, 11.67 M single-channel), 46.7 MB, 5.91 GFLOPs (5.65 single-channel), 80.8–83.6 ms per window on one CPU thread (59.8–61.9× real time), 21.3–22.9 ms with 4 threads, and 0.73–0.79 ms on the GPU (about 1,270–1,370 windows/s). Feature extraction is 8.1–9.7 ms of the single-thread time. Running Perch v2 with the probe takes 2.57 ms per window on the GPU (389 windows/s, 3.3× the ResNet GPU time) and has no CPU path.

Table 11: Quality versus cost (`bang_for_buck.csv`), sorted by single-thread CPU time. Post-hoc variants inherit the cost of `full`. "Gain/ms" is the macro-AUC change relative to `resnet_baseline` divided by the extra CPU milliseconds (undefined for the baseline itself and for the GPU-only probe). A variant is on the Pareto frontier if no other variant is both cheaper and better on macro ROC-AUC.

| Variant | Macro AUC | CPU1 ms | GPU ms | Train GPU-h | Gain/ms | Pareto (CPU) | Pareto (GPU) |
|---|---|---|---|---|---|---|---|
| `single_channel_mel` | 0.7230 | 80.82 | 0.731 | 0.438 | +0.0121 | ✓ | ✓ |
| `no_rare_upsample` | 0.6896 | 81.29 | 0.785 | 0.315 | +0.0288 | | |
| `no_soundscape_data` | 0.6656 | 81.39 | 0.784 | 0.484 | +0.0407 | | |
| `mean_freq_pool` | 0.7466 | 81.39 | 0.783 | 0.477 | +0.0047 | ✓ | |
| `no_distill` | 0.7292 | 81.54 | 0.783 | 0.485 | +0.0134 | | |
| `no_mixup` | 0.6630 | 81.55 | 0.784 | 0.320 | +0.0452 | | |
| `full` | 0.7142 | 81.92 | 0.785 | 0.463 | +0.0250 | | |
| `head_clip_only` | 0.7050 | 81.92 | 0.785 | 0.463 | +0.0304 | | |
| `full+tax_smooth` | 0.7632 | 81.92 | 0.785 | 0.463 | −0.0035 | | |
| `full+all_posthoc` | 0.8407 | 81.92 | 0.785 | 0.463 | −0.0487 | ✓ | ✓ |
| `full+site_hour_prior` | 0.7346 | 81.92 | 0.785 | 0.463 | +0.0131 | | |
| `full+gauss_smooth` | 0.7206 | 81.92 | 0.785 | 0.463 | +0.0213 | | |
| `head_framemax_only` | 0.7145 | 81.92 | 0.785 | 0.463 | +0.0248 | | |
| `no_pseudo_sc` | 0.7742 | 81.99 | 0.784 | 0.487 | −0.0104 | | ✓ |
| `no_pseudo_focal` | 0.7049 | 82.03 | 0.784 | 0.478 | +0.0325 | | |
| `mean_time_pool` | 0.6984 | 82.20 | 0.786 | 0.475 | +0.0411 | | |
| `no_specaug` | 0.7135 | 82.38 | 0.783 | 0.328 | +0.0346 | | |
| `distill_no_stopgrad` | 0.6970 | 82.39 | 0.785 | 0.320 | +0.0484 | | |
| `resnet_baseline` | 0.7571 | 83.63 | 0.782 | 0.458 | – | | ✓ |
| `perch_attn_probe` | 0.7109 | – | 2.569 | 0.156 | – | | |

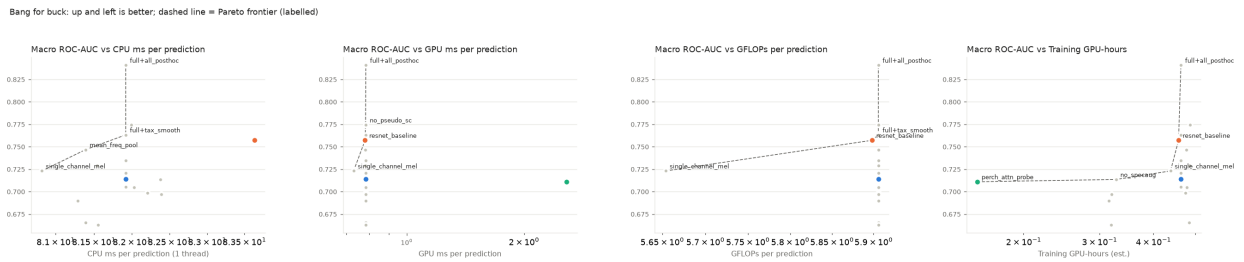



Figure 5: Quality against cost (CPU ms, GPU ms, GFLOPs and training GPU-hours per prediction). Dashed lines are the Pareto frontiers. On CPU the frontier is `single_channel_mel` → `mean_freq_pool` → `full+all_posthoc`. On GPU it is `single_channel_mel` → `resnet_baseline` → `no_pseudo_sc` → `full+all_posthoc`.

Table 12: Training cost per variant (3 folds). Three runs shared one L4 GPU at a time, so GPU-hours are estimated as wall time / 3, and epoch times depend on what the other two lanes were running.

| Variant | Wall (s) | GPU-h (est.) | Mean epoch (s) | Peak GPU mem (GB) | Epochs |
|---|---|---|---|---|---|
| `resnet_baseline` | 4942 | 0.458 | 163.9 | 2.187 | 10 |
| `perch_attn_probe` | 1689 | 0.156 | – | 0.143 | 30 |
| `full` | 5005 | 0.463 | 166.0 | 2.199 | 10 |
| `no_distill` | 5241 | 0.485 | 173.9 | 2.190 | 10 |
| `distill_no_stopgrad` | 3455 | 0.320 | 114.4 | 2.196 | 10 |
| `no_pseudo_focal` | 5165 | 0.478 | 171.3 | 2.199 | 10 |
| `no_pseudo_sc` | 5262 | 0.487 | 174.5 | 2.196 | 10 |
| `single_channel_mel` | 4735 | 0.438 | 157.0 | 2.133 | 10 |
| `mean_freq_pool` | 5147 | 0.477 | 170.7 | 2.197 | 10 |
| `mean_time_pool` | 5129 | 0.475 | 170.1 | 2.198 | 10 |
| `no_mixup` | 3451 | 0.320 | 114.3 | 2.199 | 10 |
| `no_soundscape_data` | 5224 | 0.484 | 173.3 | 2.199 | 10 |
| `no_specaug` | 3541 | 0.328 | 117.3 | 2.199 | 10 |
| `no_rare_upsample` | 3405 | 0.315 | 112.8 | 2.199 | 10 |

### 4.7 Training dynamics

Table 13: Pooled out-of-fold non-S22 macro ROC-AUC after every epoch (`all_curves.csv`). For the Perch probe (30 epochs) the first 10 epochs and the final epoch are shown. Best epoch per row in bold.

| Variant | ep0 | ep1 | ep2 | ep3 | ep4 | ep5 | ep6 | ep7 | ep8 | ep9 | ep29 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| `resnet_baseline` | 0.648 | 0.710 | 0.725 | 0.743 | 0.725 | 0.726 | 0.731 | 0.711 | 0.718 | **0.761** | – |
| `perch_attn_probe` | 0.679 | 0.693 | 0.703 | 0.705 | 0.703 | 0.705 | 0.715 | 0.703 | 0.711 | 0.711 | 0.711 |
| `full` | 0.512 | 0.452 | 0.495 | 0.552 | 0.634 | 0.662 | 0.690 | 0.677 | 0.697 | **0.713** | – |
| `no_distill` | 0.562 | 0.617 | 0.684 | 0.690 | 0.678 | 0.715 | 0.717 | 0.709 | **0.732** | 0.680 | – |
| `distill_no_stopgrad` | 0.465 | 0.640 | 0.673 | 0.680 | 0.655 | 0.692 | 0.696 | 0.664 | 0.686 | **0.709** | – |
| `no_pseudo_focal` | 0.519 | 0.448 | 0.500 | 0.519 | 0.600 | 0.655 | 0.682 | 0.679 | 0.698 | **0.707** | – |
| `no_pseudo_sc` | 0.463 | 0.517 | 0.618 | 0.694 | 0.715 | 0.723 | 0.749 | 0.760 | 0.755 | **0.764** | – |
| `single_channel_mel` | 0.459 | 0.434 | 0.476 | 0.566 | 0.643 | 0.674 | 0.704 | 0.712 | **0.723** | 0.719 | – |
| `mean_freq_pool` | 0.492 | 0.487 | 0.521 | 0.552 | 0.620 | 0.647 | 0.696 | 0.705 | 0.722 | **0.747** | – |
| `mean_time_pool` | 0.498 | 0.462 | 0.482 | 0.494 | 0.572 | 0.608 | 0.646 | 0.669 | 0.686 | **0.706** | – |
| `no_mixup` | 0.512 | 0.449 | 0.468 | 0.452 | 0.516 | 0.555 | 0.597 | 0.606 | 0.639 | **0.677** | – |
| `no_soundscape_data` | 0.455 | 0.495 | 0.558 | 0.559 | 0.552 | 0.611 | 0.606 | 0.652 | 0.673 | **0.704** | – |
| `no_specaug` | 0.494 | 0.531 | 0.507 | 0.599 | 0.640 | 0.682 | 0.679 | 0.702 | **0.710** | 0.710 | – |
| `no_rare_upsample` | 0.500 | 0.441 | 0.491 | 0.505 | 0.587 | 0.616 | 0.647 | 0.666 | 0.681 | **0.701** | – |

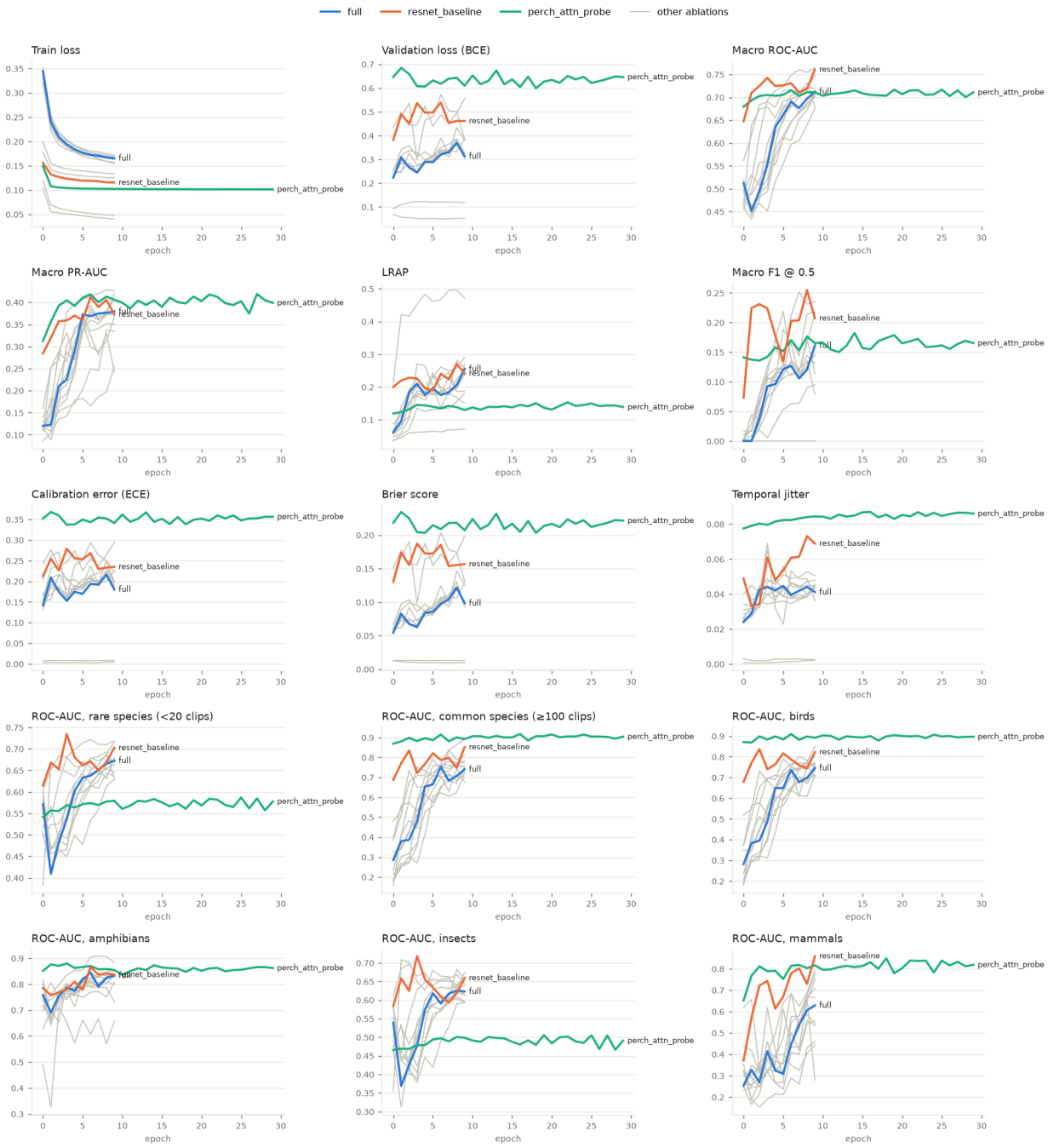



Figure 6: Pooled learning curves for all metrics. `full` (blue), `resnet_baseline` (orange) and `perch_attn_probe` (green) are highlighted; grey lines are the other ablations.

Four patterns stand out (Table 13, Figure 6):

1. **Slow start of the distilled models.** `full`'s pooled AUC is 0.512, 0.452 and 0.495 in epochs 0–2, below or near chance, and then rises steadily to 0.713 at epoch 9. It is *still rising at the end of the budget*. The same shape appears for every variant with stop-gradient distillation.

2. **Fast start without stop-gradient.** `resnet_baseline` reaches 0.710 after two epochs

(epoch 1) and ends at 0.761. `no_distill` reaches 0.684 by epoch 2, peaks at 0.732 (epoch 8) and falls to 0.680 at epoch 9. `distill_no_stopgrad` reaches 0.640 at epoch 1.

3. **Perch probe plateau.** The probe is at 0.679 after one epoch and then fluctuates between 0.693 and 0.717 for 29 more epochs.

4. **Calibration drift.** `full`'s validation BCE rises from 0.22 to 0.31–0.37 while AUC improves, and its ECE stays around 0.15–0.22. The variants without pseudo-labelled soundscapes (`no_pseudo_sc`, `no_soundscape_data`) keep validation BCE near 0.05–0.12 and ECE below 0.01 throughout. Training loss follows the same split: 0.35→0.17 for `full` versus 0.12→0.05 for `no_pseudo_sc`.

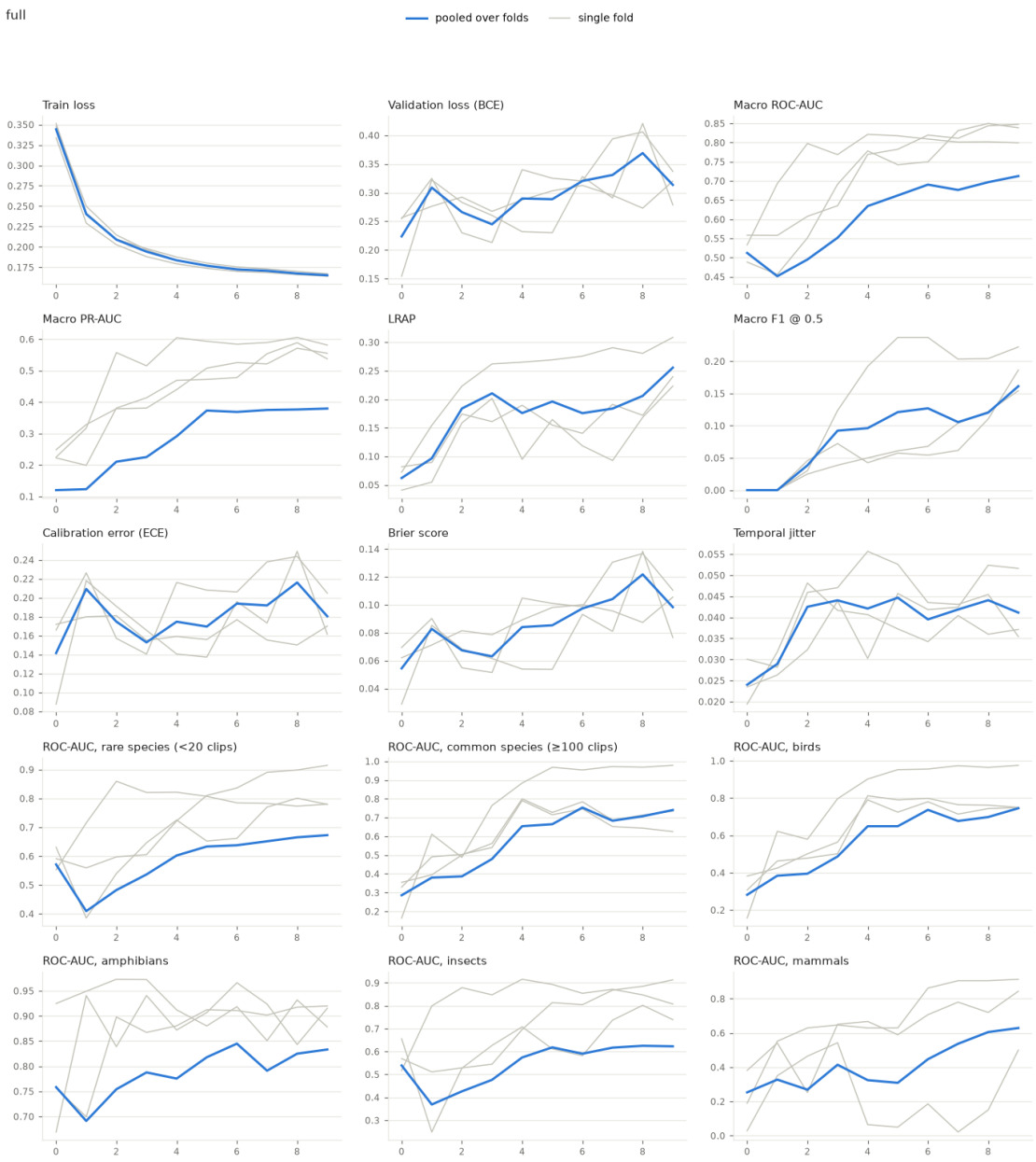
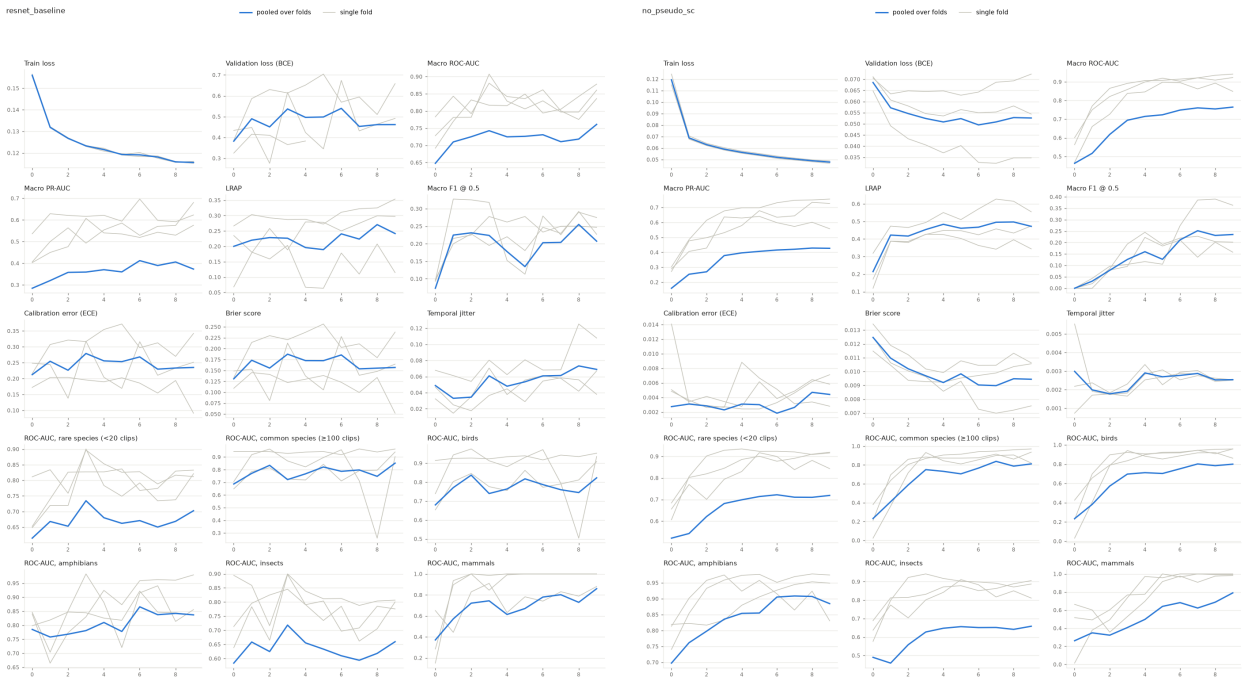



Figure 7: `full`: per-fold (grey) and pooled (blue) curves for every metric. Fold-local AUC (0.80–0.85 at the end) is well above the pooled curve.




(a) `resnet_baseline` (b) `no_pseudo_sc`

Figure 8: Per-fold and pooled curves for the two strongest trained models.

## 4.8 Attention behaviour

For 16 fixed labelled non-S22 validation windows per fold, the attention weights over time and the class-activation map (the classifier applied to every cell of the feature map) for the window's true species were recorded after every epoch. We analysed the saved fold-0 tensors at the epoch with the best fold-local non-S22 AUC. The fold-0 windows cover five classes (insect sonotypes `47158son13` and `47158son10`, the caiman `116570`, `516975` and the frog `1491113`).

- **`full` (epoch 4).** The attention over the 10 time steps is close to uniform: the mean maximum weight is 0.120 against 0.100 for uniform, and the mean normalised entropy is 0.962. The average profile is $[0.028, 0.039, 0.116, 0.116, 0.115, 0.115, 0.116, 0.117, 0.118, 0.120]$, so the first second is consistently down-weighted and the rest is flat. Profiles are similar across windows of different species (mean pairwise correlation 0.585). The predicted probability of the true species at that epoch averages 0.068 (median 0.027; range 0.012–0.251), up from 0.035 at epoch 0.

- **`resnet_baseline` (epoch 3).** By construction the attention is uniform (normalised entropy 1.0). Its probability for the true species averages 0.321 (median 0.177), up from 0.144 at epoch 0.

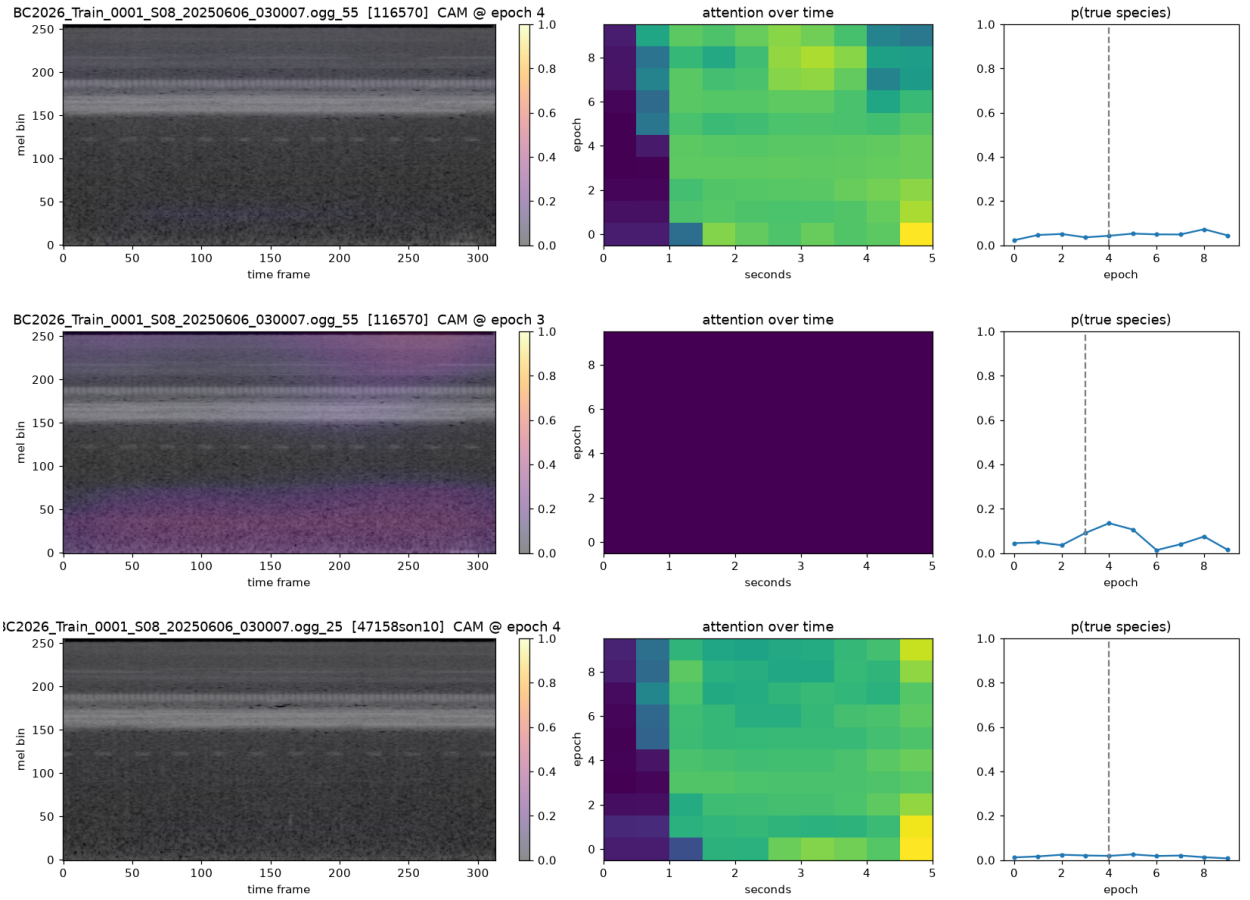



Figure 9: Tracked windows (left: log-mel with class-activation overlay; middle: attention over time per epoch; right: probability of the true species per epoch, dashed line = best epoch). Top: `full` on a caiman (`116570`) window. Middle: `resnet_baseline` on the same window. Bottom: `full` on an insect sonotype (`47158son10`). The `full` attention has the same position-driven shape for both species: low for the first second, highest in the last 0.5 s at early epochs, and flat in between.

# 5 Interpretation

## 5.1 How much can these rankings be trusted?

The study is carefully controlled, but small. The primary metric averages 44 classes, evaluated on 16 files from 8 sites, and 26 of those classes are rare. A single file can shift a class's AUC a lot. The bootstrap intervals (Table 6) are wide relative to most of the ablation effects, so we group findings by strength:

- **Robust:** the macro-AUC effects whose bootstrap interval excludes zero (mixup helps, mean beats GeM frequency pooling), and effects that are huge on metrics other than AUC and fit a clear mechanism. The latter are the calibration and ranking harm from pseudo-labelled soundscapes (ECE 0.184 vs 0.003, LRAP 0.221 vs 0.477), the collapse of detection without soundscape data (F1 0, PR-AUC −0.176), the post-hoc gains, and the cost equivalence of all CNN variants (a measurement, not a sample).

- **Suggestive:** effects of 0.015–0.06 AUC that point in a consistent direction but whose interval just includes zero: the gain from removing pseudo-labelled soundscapes, the plain baseline beating `full`, and attention versus mean time pooling.
- **Noise-level:** effects under about 0.01 AUC (SpecAugment, single versus delta channels, rare upsampling on rare classes) and anything carried mainly by **Mammalia**. Mammal AUC ranges from 0.4066 to 0.8447 across variants that differ only slightly, and its per-fold value swings between 0.02 and 0.91 from epoch to epoch.

Two further caveats apply. First, checkpoints were chosen on the same validation folds that are scored, which inflates every absolute number (although it affects all variants alike). Second, every variant got the same short budget of 6,000 steps, which favours components that learn quickly over ones that learn well eventually.

### 5.2 Pseudo-labelled soundscapes are the main problem

Removing the 24,000 pseudo-labelled soundscape windows is the single best training change: macro AUC +0.060 (better in 97.4% of bootstrap resamples), PR-AUC +0.069, LRAP +0.256, F1 +0.106, and ECE from 0.184 to 0.003. The pre-registered hypothesis expected the opposite, on the grounds that unlabelled soundscapes are in-domain. We see three reasons it went the other way.

1. **The soft targets are not probabilities of presence.** Fused Perch scores plus site/hour prior logits give many classes moderate probabilities in every window. One batch in five trains the student to reproduce those values. The result is a model that predicts moderately high probabilities everywhere: validation BCE of 0.22–0.37 and ECE of 0.15–0.22, against 0.05–0.12 BCE and ECE below 0.01 without that stream. The training loss of `full` levels off near 0.17 because soft targets have irreducible BCE.
2. **The teacher is not much better than the student.** The Perch probe reaches only 0.711 macro AUC on these soundscapes and 0.578 on rare species. Pseudo-labels cannot carry knowledge the teacher does not have, especially for the 28 classes with no focal clips and for sonotypes with no Perch mapping, which receive the prior alone or $-8$.
3. **Ranking suffers with calibration.** LRAP, which ranks species within each window, nearly doubles without the stream (0.221 $\rightarrow$ 0.477). That fits a picture in which the prior-dominated soft targets push the same site-typical species to the top of every window regardless of what is calling.

### 5.3 Distillation with stop-gradient slows learning

In `full` the backbone never sees a classification gradient: it learns only to imitate Perch embeddings, and a small head learns to read species from those features. With 6,000 steps this is slow. Pooled AUC stays at or below chance for three epochs and is still rising at epoch 9. Both alternatives that let classification reach the backbone start much faster (`resnet_baseline`, `no_distill`). The end result is mixed: `no_distill` is +0.015 AUC and +0.033 rare AUC better than `full`, but worse calibrated (ECE +0.041, which supports that hypothesis). `distill_no_stopgrad` is $-0.017$ AUC and $-0.119$ on common species, yet has a much higher F1 (+0.101). Our reading is that distillation adds a *regulariser and a delay*. At this budget the delay dominates, and the cheaper end-to-end models win. Whether the distilled student overtakes them with the notebook's 25 full epochs is the most important open question. It also matters for cost, because distillation is free at inference.

### 5.4 Simple beats sophisticated at this budget

The pooled ResNet-18 baseline, with no attention, no GeM and no distillation, is the second-best trained model (0.7571). It is better than `full` on every taxon except Amphibia (−0.010), on rare (+0.042) and common (+0.073) species, on LRAP (+0.056) and on F1 (+0.118). It is worse on calibration (ECE 0.226) and jitter (0.068), because it also trains on the pseudo-labelled stream. The SED-head ablations point the same way:

- **GeM versus mean frequency pooling.** Mean pooling is better (+0.032 AUC, +0.084 F1, +0.069 LRAP), one of only two ablations whose bootstrap interval excludes zero. The prediction that insects and frogs, being narrow-band, would need GeM was not supported (Insecta +0.015 with mean pooling).
- **Attention versus mean time pooling.** Attention helps a little (mean pooling gives −0.016 AUC and −0.011 LRAP, supporting the hypothesis). The attention maps explain why the gain is small: they are nearly uniform and mostly encode *position*, down-weighting the first second of every window (Section 4.8). The position effect may come from zero-padding at the window edge in the convolutional receptive field. The model is not yet acting as an event detector.
- **Clip versus frame-max heads.** Frame-max alone matches the blend (0.7145 versus 0.7142) and is more over-confident (ECE +0.036, supported). Clip-only is worse (0.7050). The 50/50 blend mainly buys calibration.

### 5.5 Delta channels are not worth it

Dropping the delta and delta-delta channels changes macro AUC by +0.009, which is within noise, and Insecta by only −0.002. That technically "supports" the hypothesis, but the effect is negligible. It also gives the cheapest model (5.65 versus 5.91 GFLOPs, 0.73 versus 0.79 ms on GPU, 80.8 versus 81.9 ms on CPU) and the start of every Pareto frontier. The idea that temporal derivatives help with repetition-rate-defined insect and frog calls is not supported. A ResNet's first layers can learn such derivatives from log-mel alone.

### 5.6 Mixup and labelled soundscapes are essential

Removing mixup costs the most AUC of any training change (−0.051, the only harmful change whose bootstrap interval excludes zero) and lowers every taxon. Removing soundscape data costs −0.049 AUC, −0.176 PR-AUC, −0.162 LRAP, F1 down to zero, Amphibia −0.257 and rare −0.086. Yet it *raises* common-species AUC (+0.102). Frogs and insects are recorded mainly in soundscapes, and soundscape backgrounds differ greatly from focal recordings. A model trained only on focal clips stays good at ranking common birds but is not calibrated for the deployment domain: no probability crosses 0.5. Focal–soundscape mixup and the small labelled-soundscape stream (about 630 windows per fold) are what adapt the model to the field.

### 5.7 The free wins: post-processing

Every post-hoc step that uses context the network cannot see helped, and they stack:

- **Site/hour prior:** AUC +0.020, LRAP +0.175, ECE 0.184 → 0.006, Mammalia +0.172. Where and when a recording was made strongly constrains which species can be present.

- **Taxonomy smoothing:** rare +0.063, Amphibia +0.073, Insecta +0.082. Rare species borrow evidence from relatives in the same genus.
- **Temporal smoothing:** jitter 0.044 → 0.012 and PR-AUC +0.031. Calls last longer than 5 s and soundscape choruses persist.
- **All three:** AUC 0.8407 (+0.127), rare 0.8335, Insecta 0.8558, Amphibia 0.9139, jitter 0.0013, ECE 0.0062. The "free" lookup and averaging steps contribute more than any architectural component.

One caution applies. Folds are grouped by file, not by site, so the prior for a validation file is fit partly on other recordings from the *same* sites. The gain therefore reflects deployment at sites with labelled history. At a brand-new site the site and site-hour terms shrink back to the hourly and global priors, and the benefit would be smaller.

### 5.8 The teacher alone: good on birds, poor elsewhere, expensive

The Perch probe is the best ranker of common species (0.904) and birds (0.897), as expected from a model trained on huge bird collections. It is the worst on rare species (0.578, against a predicted rise), insects (0.491), calibration (ECE 0.356) and jitter, and second-worst on LRAP (0.139). Its curve plateaus after one epoch, so the frozen features, not the training budget, limit it. It also needs a GPU (2.57 ms per window, no CPU path). For a CPU-only field station it is not an option, and its knowledge reaches the student only where it is strong (birds).

### 5.9 Cost: the architecture choices are free

The most practical result is that *none of the studied components change inference cost meaningfully.* All CNN variants differ by less than 4% in CPU time (80.8–83.6 ms per window, single thread) and by less than 8% in GFLOPs. Distillation, pseudo-labels, mixup, SpecAugment and upsampling are training-time only, and the post-hoc steps are table lookups and averages. One CPU thread processes a 5 s window in about 82 ms, i.e. 61 seconds of audio per second. Four threads take about 22 ms per window, roughly 227× real time, which is ample headroom for many recorders per box. The cost question is decided by the backbone (ResNet-18 here), and every component choice should be made purely on quality.

### 5.10 Pooled versus fold-local scoring

Fold-local AUCs (0.70–0.94) are far above the pooled out-of-fold scores (0.66–0.77). Part of the gap is selection bias, since each checkpoint was picked on its own fold. The rest comes from pooling: the three fold models produce scores on different scales, and a single ranking across them penalises that inconsistency. Deployment uses one model with one score scale, so the pooled number is closer to deployment behaviour for threshold-free ranking. Even so, the per-fold numbers show that within one model the ranking of species is considerably better than the pooled score suggests.

### 5.11 What the failed hypotheses teach

The 13 supported hypotheses mostly concern cheap, well-understood components: mixup, soundscape data, SpecAugment, time attention, smoothing and priors. The 8 unsupported ones are distillation helping rare species, pseudo-labelled soundscapes helping, Perch beating the student on rare species, GeM helping narrow-band taxa, the pooled baseline being worse than the SED head,

the clip/frame blend beating either head, Perch-localised focal targets helping precision, and stop-gradient being harmless. Five of the eight are about using the teacher (distillation, stop-gradient, Perch-localised targets, pseudo-labelled soundscapes, Perch on rare species): every idea that relies on the teacher's knowledge failed to pay off at this budget.

# 6 Limitations

1. **Small evaluation set:** 192 windows, 16 files, 8 sites, 44 scorable classes and a single seed. Mammalia and other low-count groups are especially unstable.
2. **Optimistic selection:** the checkpoint epoch is chosen on the scored validation fold, using a criterion (all windows, S22 included) that differs from the reported metric (non-S22).
3. **Short budget:** 10×600 steps, constant learning rate, and only 3 of 5 folds trained. Slower-learning components are disadvantaged, and `full` had not converged.
4. **One change at a time:** interactions were not tested. In particular the obvious combination (no pseudo-soundscapes, mean frequency pooling, single channel, plus post-processing) was not trained.
5. **Site leakage in priors:** site/hour priors and pseudo-label fusion use other files from the same sites.
6. **Shared-GPU timing:** training times were measured with three concurrent runs, so epoch times reflect lane contention. Inference timings were measured on an idle machine.
7. **Augmentation change:** waveform gain and noise augmentation from the original notebook were dropped when spectrograms were precomputed.

# 7 Recommendations

For a CPU-only on-site monitor, the evidence supports:

1. A ResNet-18 trained end-to-end on single-channel log-mel, with the classification gradient reaching the backbone, mean frequency pooling, and the attention or pooled head.
2. Training *with* mixup (focal–focal and focal–soundscape) and labelled soundscapes, and *without* the pseudo-labelled soundscape stream, at least until its targets are recalibrated (for example by sharpening, thresholding or confidence weighting).
3. The three post-processing steps (site/hour prior, temporal smoothing, taxonomy smoothing), which cost nothing to run and gave the largest single gain in the study.
4. Before deployment, confirming the combined configuration with all 5 folds, several seeds, the full training budget and site-grouped validation. This would also settle whether distillation pays off once training is long enough.

Such a model runs at about 61× real time on one CPU thread.

# 8 Conclusion

We set out to find the cheapest model that still performs well for continuous, on-site acoustic monitoring of the Pantanal's birds, frogs, insects and mammals. Under a fixed, short training budget, the elaborate distilled SED network (0.7142 macro ROC-AUC) was beaten by a plain ResNet-18 (0.7571) and by the same network without pseudo-labelled soundscapes (0.7742). The latter also cut calibration error by a factor of about 57. Mixup and labelled soundscapes are essential. Delta channels, GeM pooling and Perch-based supervision did not earn their place. Every component is free at inference (about 82 ms per window on one CPU thread), so the choice should be made on quality. Free post-processing with site/hour priors, temporal smoothing and taxonomy smoothing raised macro ROC-AUC to 0.8407. With an evaluation set of 16 files, the large effects are trustworthy and the small ones are not. The clear next step is to train the combined simple configuration longer, on all folds, with site-grouped validation.


# Acknowledgements and Attribution

The SED head, Perch distillation with stop-gradient, pseudo-labelling system, site/hour priors and taxonomy smoothing are adapted from nina2025's Kaggle notebook "BirdCLEF 2026 EoS.9" [1]. That notebook builds on tuckerarrants' distilled-SED and waveform-cache datasets and the BirdCLEF 2025 first-place SED design by nikitababich. Perch v2 is by Google DeepMind/Google Research [3]. Data are from the BirdCLEF 2026 competition [2].


# References


[1] nina2025. *BirdCLEF 2026 EoS.9* (Kaggle notebook, sections Model_1 and Model_22). `https://www.kaggle.com/code/nina2025/birdclef-2026-eos-9`

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